



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

PRE - SCORE

14-10-2022

## JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S(TEAS)]

ANSWER KEY

PAPER-1

### PART-1 : PHYSICS

|                 |    |         |         |       |       |      |         |
|-----------------|----|---------|---------|-------|-------|------|---------|
| SECTION-I (i)   | Q. | 1       | 2       | 3     | 4     |      |         |
|                 | A. | D       | D       | B     | B     |      |         |
| SECTION-I (ii)  | Q. | 5       | 6       | 7     | 8     | 9    | 10      |
|                 | A. | A,B,C,D | A,B,C,D | A,B   | A,B,C | B,C  | A,B,C,D |
| SECTION-II (i)  | Q. | 1       | 2       | 3     | 4     | 5    | 6       |
|                 | A. | 25.00   | 975.00  | 50.00 | 50.00 | 2.40 | 80.00   |
| SECTION-II (ii) | Q. | 7       | 8       | 9     |       |      |         |
|                 | A. | 20      | 2       | 9     |       |      |         |

### PART-2 : CHEMISTRY

|                 |    |       |       |         |      |        |         |
|-----------------|----|-------|-------|---------|------|--------|---------|
| SECTION-I (i)   | Q. | 1     | 2     | 3       | 4    |        |         |
|                 | A. | D     | A     | C       | A    |        |         |
| SECTION-I (ii)  | Q. | 5     | 6     | 7       | 8    | 9      | 10      |
|                 | A. | A,B,C | A,C,D | A,B,C,D | B,D  | A,B,C  | A,B,C,D |
| SECTION-II (i)  | Q. | 1     | 2     | 3       | 4    | 5      | 6       |
|                 | A. | 7.00  | 3.00  | 6.00    | 9.00 | 100.00 | 12.00   |
| SECTION-II (ii) | Q. | 7     | 8     | 9       |      |        |         |
|                 | A. | 1     | 5     | 4       |      |        |         |

### PART-3 : MATHEMATICS

|                 |    |       |        |      |       |        |         |
|-----------------|----|-------|--------|------|-------|--------|---------|
| SECTION-I (i)   | Q. | 1     | 2      | 3    | 4     |        |         |
|                 | A. | B     | D      | D    | A     |        |         |
| SECTION-I (ii)  | Q. | 5     | 6      | 7    | 8     | 9      | 10      |
|                 | A. | B,C,D | A,B    | B,C  | A,C,D | A,B    | A,B,C,D |
| SECTION-II (i)  | Q. | 1     | 2      | 3    | 4     | 5      | 6       |
|                 | A. | 32.00 | 108.00 | 9.00 | 0.50  | 200.00 | 100.00  |
| SECTION-II (ii) | Q. | 7     | 8      | 9    |       |        |         |
|                 | A. | 9     | 8      | 329  |       |        |         |

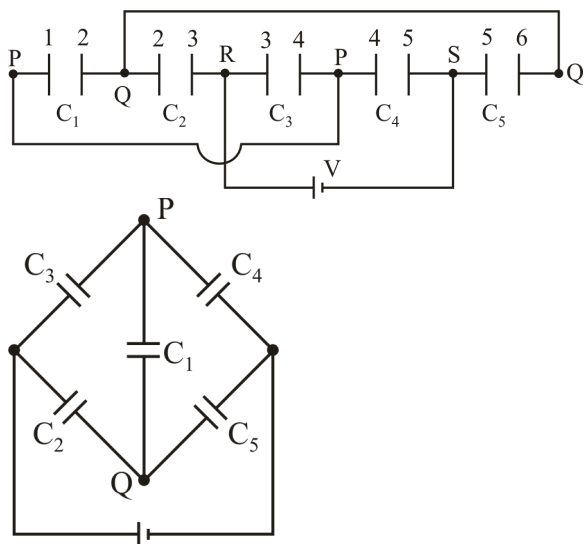
**HINT – SHEET**

## PART-1 : PHYSICS

### SECTION-I (ii)

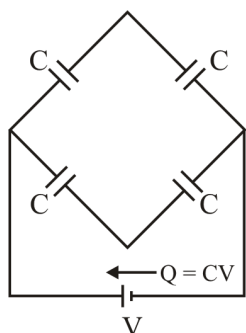
6. Ans ( A,B,C,D )

(A)



$$C_1 = C_2 = C_3 = C_4 = C_5 = C = \frac{\epsilon_0 A}{d}$$

(B) Balanced wheat stone bridge



$$W_B = CV^2 = \frac{\epsilon_0 A}{d} V^2$$

Electric field between 3 & 4

$$= E_3 = \frac{Q_3}{A\epsilon_0} = \frac{\frac{\epsilon_0 A}{d} \frac{V}{2}}{A\epsilon_0} = \frac{V}{2d}$$

Potential difference between plate 2 and 5

$$= V_Q - V_S = \frac{V}{2}$$

Potential difference between plate 1 and 4

$$= V_P - V_S = \frac{V}{2}$$

8. Ans ( A,B,C )

$$\vec{F} = q(x\hat{i} + y\hat{j}) \times (y\hat{i} + x\hat{j})$$

$$= q(x^2\hat{k} - y^2\hat{k})$$

$$= q(x^2 - y^2)\hat{k}$$

## PART-1 : PHYSICS

### SECTION-II (i)

2. Ans ( 975.00 )

The fan is operating at 200 V, consuming 1000 W,

$$\text{then } I = \frac{1000}{200} = 5A$$

But as coil resistance is  $1\Omega$  then power dissipated by internal resistance heat is  $P_1 = I^2 R = 25W$

If V is net emf across coil then

$$\frac{V^2}{R} = 25W ; V = 5 \text{ volt}$$

Net emf = source emf – back emf

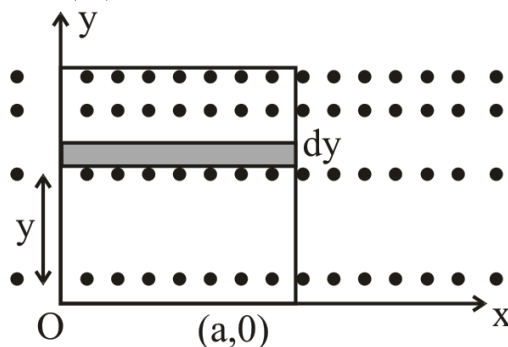
$$V = V_S - e \Rightarrow e = 195 V$$

The work done  $P_2 = 1000 - 25 = 975 W$ .

## PART-1 : PHYSICS

### SECTION-II (ii)

8. Ans ( 2 )



$$\phi = \int \vec{B} \cdot d\vec{A}$$

$$\phi = \int cy^3 t^2 a dy$$

$$\phi = ct^2 a \cdot \int_0^a y^3 dy$$

$$\phi = ct^2 a \cdot \frac{a^4}{4}$$

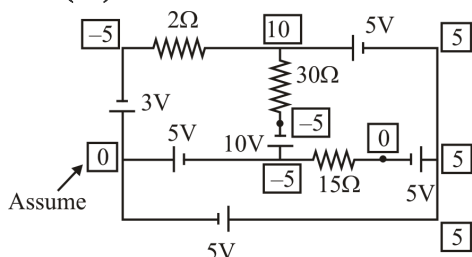
$$\phi = \frac{ct^2 a^5}{4}$$

$$e = \left| -\frac{d\phi}{dt} \right|$$

Now induced emf

$$e = \left| -\frac{2cta^5}{4} \right| = \frac{2cta^5}{4} = \frac{cta^5}{2}$$

9. Ans ( 9 )



Potentials are indicated in figure

Current in  $2\Omega = \frac{10 - (-5)}{2} = \frac{15}{2} = 7.5A$ ,  
leftwards

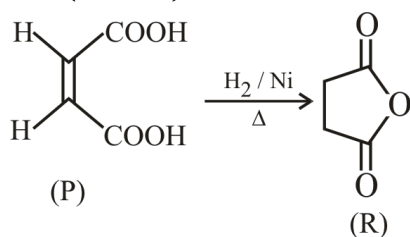
Current in  $30\Omega = \frac{10 - (-15)}{30} = \frac{25}{30} = \frac{5}{6}A$ ,  
downwards

$\frac{i_1}{i_2} = 9$  Ans.

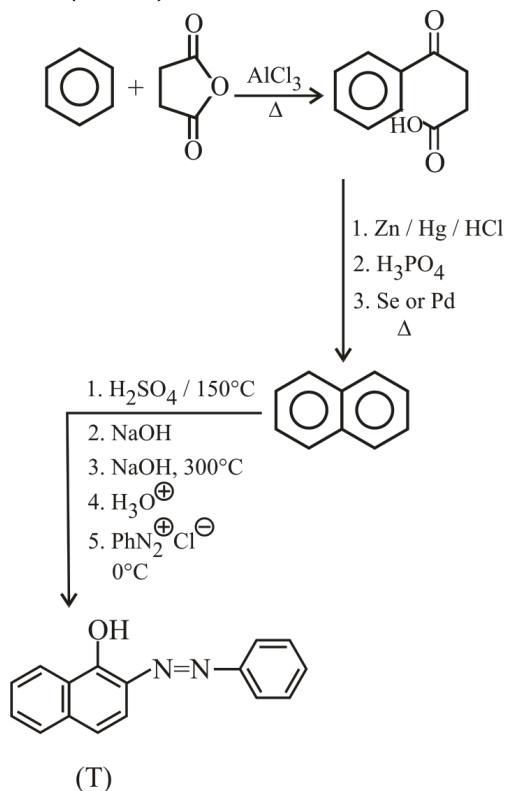
## PART-2 : CHEMISTRY

### SECTION-II (i)

5. Ans ( 100.00 )



6. Ans ( 12.00 )



## PART-3 : MATHEMATICS

### SECTION-I (i)

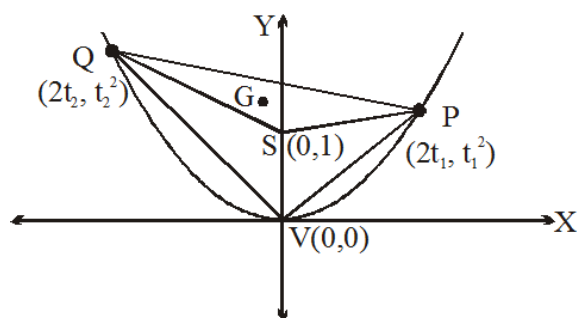
1. Ans ( B )

Let  $P(2t_1, t_1^2)$ ;  $Q(2t_2, t_2^2)$

Clearly,  $t_1 t_2 = -4$  .....(1)

Also,  $h = \frac{2t_1 + 2t_2}{3}$  .....(2)

$\Rightarrow \frac{3h}{2} = t_1 + t_2$



and  $k = \frac{1 + t_1^2 + t_2^2}{3}$  .....(3)

$\Rightarrow 3k = 1 + t_1^2 + t_2^2$

$\therefore$  On eliminating we get  $t_1, t_2$  from (1), (2)

and (3) ,

$$\frac{9h^2}{4} = 3k - 9$$

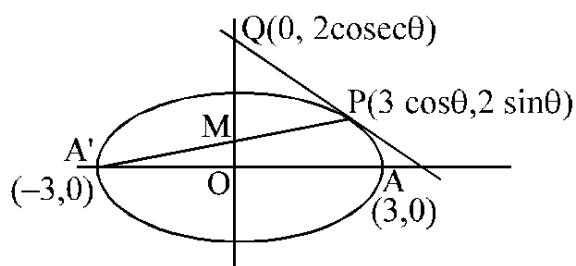
$$\Rightarrow x^2 = \frac{4}{3} (y - 3).$$

2. Ans (D)

$$a = 3 ; b = 2$$

$$T : \frac{x \cos \theta}{3} + \frac{y \sin \theta}{2} = 1$$

$$x = 0 ; y = 2 \operatorname{cosec} \theta$$



$$\text{chord } A'P, y = \frac{2 \sin \theta}{3(\cos \theta + 1)}(x + 3)$$

$$\text{put } x = 0 \quad y = \frac{2 \sin \theta}{1 + \cos \theta} = OM$$

$$\text{Now } OQ^2 - MQ^2 = OQ^2 - (OQ - OM)^2$$

$$= 2(OQ)(OM) - OM^2 = OM \{ 2(OQ) - (OM) \}$$

$$= \frac{2 \sin \theta}{1 + \cos \theta} \left[ \frac{4}{\sin \theta} - \frac{2 \sin \theta}{1 + \cos \theta} \right]$$

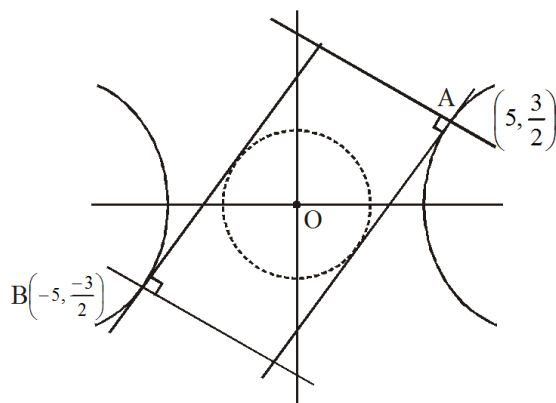
$$= \frac{4 \sin \theta}{1 + \cos \theta} \left[ \frac{2(1 + \cos \theta) - (1 - \cos^2 \theta)}{\sin \theta(1 + \cos \theta)} \right]$$

$$= \frac{4(1 + \cos \theta)(2 - 1 + \cos \theta)}{(1 + \cos \theta)(1 + \cos \theta)} = 4$$

3. Ans (D)

$$\frac{x^2}{16} - \frac{y^2}{4} = 1$$

$\left(5, \frac{3}{2}\right)$  lies on the curve.



tangent at A

$$\frac{5x}{16} - \frac{3}{2} \left( \frac{y}{4} \right) = 1 \Rightarrow 5x - 6y = 16$$

tangent at B

$$5x - 6y = -16$$

Normal at A

$$6x + 5y = \lambda$$

$$A \left( 5, \frac{3}{2} \right) \quad 30 + \frac{15}{2} = \lambda$$

$$\Rightarrow \lambda = \frac{75}{2} \quad 6x + 5y = \frac{75}{2}$$

$$\text{||}^y \text{ normal at B is } 6x + 5y = \frac{-75}{2} \text{ distance}$$

$$\text{between tangents is } 2r = \frac{32}{\sqrt{25 + 36}}$$

$$\therefore r = \frac{16}{\sqrt{61}}$$

Radius of circumcircle of rectangle is

$$R = \sqrt{25 + \frac{9}{4}} = \frac{\sqrt{109}}{2}. \text{ Ans.}$$



8. Ans (A,C,D)

$$\frac{4x}{\cos \theta} - \frac{2y}{\sin \theta} = 12$$

$$A(3\cos\theta, 0) \quad B(0, -6\sin\theta)$$

$$2h = 3\cos\theta, \quad 2k = -6\sin\theta$$

$$\Rightarrow \frac{4x^2}{9} + \frac{4y^2}{36} = 1$$

$$4x^2 + y^2 = 9$$

$$(C) \frac{x^2}{16} + \frac{y^2}{4} = 1, e = \sqrt{1 - \frac{4}{16}} = \frac{\sqrt{3}}{2}$$

$$F_1(2\sqrt{3}, 0) \quad \& \quad F_2(-2\sqrt{3}, 0)$$

$$B_1(0, 2) \quad \& \quad B_2(0, -2)$$

$$A = \frac{1}{2} \times 4\sqrt{3} \times 4 = 8\sqrt{3}$$

$$(D) \frac{x \cos \theta}{4} + \frac{y \sin \theta}{2} = 1$$

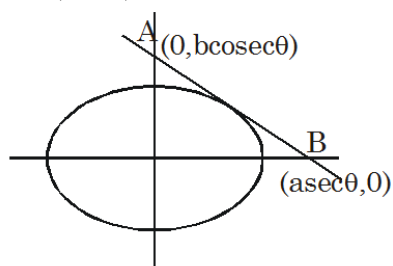
$$L = (4\sec\theta, 0) \quad M = (0, 2\csc\theta)$$

$$LM = \sqrt{16\sec^2\theta + 4\csc^2\theta}$$

$$= \sqrt{20 + 16\tan^2\theta + 4\cot^2\theta}$$

$$LM_{\min} = 6$$

9. Ans (A,B)



$$2a = 4, \quad 2b = 2\sqrt{3}$$

$$PS + PS' = 4$$

Now apply AM  $\geq$  GM

$$\frac{PS + PS'}{2} \geq (PS \cdot PS')^{1/2}$$

$$\Rightarrow PS \cdot PS' \leq 4$$

$$\frac{PS^2 + PS'^2}{2} \geq \left( \frac{PS + PS'}{2} \right)^2$$

$$\frac{PS^2 + PS'^2}{2} \geq 4 \Rightarrow PS^2 + PS'^2 \geq 8$$

$$(C) D = \sqrt{a^2 \sec^2\theta + b^2 \csc^2\theta} = AB$$

$$D_{\min} = a + b = 2 + \sqrt{3}$$

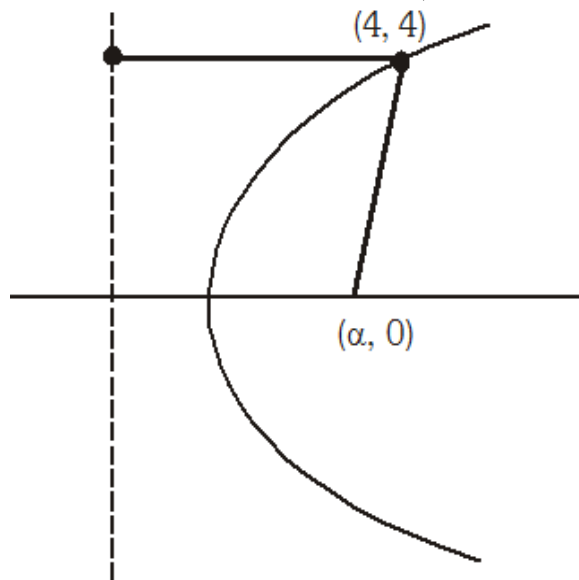
$$(D) PQ = \frac{4}{\sqrt{a^2 \sec^2\theta + b^2 \csc^2\theta}}$$

$$\Rightarrow PQ_{\max} = \frac{4}{a+b} = 8 - 4\sqrt{3}$$

10. Ans (A,B,C,D)

$$SP = PM$$

$$\sqrt{(4-\alpha)^2 + (4-0)^2} = \frac{|4\ell + 4m - 1|}{\sqrt{\ell^2 + m^2}}$$



$$\ell x + my = 1$$

$$\Rightarrow (\ell^2 + m^2)((\alpha - 4)^2 + 16) = (4(\ell + m) - 1)^2$$

$$\Rightarrow (\ell^2 + m^2)(\alpha^2 + 32 - 8\alpha) = 16(\ell + m)^2 + 1 - 8(\ell + m)$$

$$\Rightarrow (\ell^2 + m^2)(\alpha^2 + 32 - 8\alpha - 16) + 8(\ell + m) - 32\ell m = 1$$

$$\Rightarrow (\ell^2 + m^2)(\alpha - 4)^2 + 8(\ell + m) - 32\ell m - 1 = 0$$

But given that

$$(\ell^2 + m^2) + 8(\ell + m) - 32\ell m - 1 = 0$$

$$\therefore \alpha = 5 \quad \text{or} \quad 3$$

Take focus (3, 0)

Equation of directrix  $\ell x + my = 1$  satisfies (0, 3)

i.e. mirror image of

focus through tangent  $y = x$

$$0 + 3m = 1$$

slope of line joining (0, 3) and (4, 4) is  $\frac{1}{4}$  which

is parallel to axis.

$$\therefore \text{Equation of axis } y - 0 = \frac{1}{4}(x - 3)$$

$$\Rightarrow x - 4y - 3 = 0$$

## PART-3 : MATHEMATICS

### SECTION-II (i)

1. **Ans ( 32.00 )**

Chord subtend right angle at (0, 0) it means it passes through P(4,0)

$$tx + y = 2t + t^3 \xrightarrow{(4,0)} 4t = 2t + t^3 \Rightarrow t^3 = 2t$$

$$t = 0, \sqrt{2}, -\sqrt{2} \quad A, B, C \equiv (t^2, 2t)$$

$$A(0, 0), B(2, 2\sqrt{2}), C(2, -2\sqrt{2})$$

$$A = \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ 2 & 2\sqrt{2} & 1 \\ 2 & -2\sqrt{2} & 1 \end{vmatrix} = \frac{8\sqrt{2}}{2} = 4\sqrt{2}$$

2. **Ans ( 108.00 )**

Chord subtend right angle at (0, 0) it means it passes through P(4,0)

$$tx + y = 2t + t^3 \xrightarrow{(4,0)} 4t = 2t + t^3 \Rightarrow t^3 = 2t$$

$$t = 0, \sqrt{2}, -\sqrt{2} \quad A, B, C \equiv (t^2, 2t)$$

$$A(0, 0), B(2, 2\sqrt{2}), C(2, -2\sqrt{2})$$

$$t_1 = \sqrt{2}, t_2 = -t_1 - \frac{2}{t_1} \Rightarrow t_2 = -2\sqrt{2}$$

$$B(2, 2\sqrt{2}) \text{ and } B'(8, -4\sqrt{2})$$

$$BB' = \sqrt{6^2 + (6\sqrt{2})^2} = 6\sqrt{3}$$

3. **Ans ( 9.00 )**

Solve curves we get

$$y^2 = \frac{144 - 9r^2}{7}$$

$$x^2 = \frac{16r^2 - 144}{7}$$

The equation of tangent to ellipse is

$$y = mx \pm \sqrt{16m^2 + 9}$$

which is also tangent to given circle so

$$r^2 m^2 + r^2 = 16m^2 + 9 \quad \dots\dots(i)$$

Also if A'B'C'D' is square then  $m = \pm 1$

$$\therefore r = \frac{5}{\sqrt{2}}$$

4. **Ans ( 0.50 )**

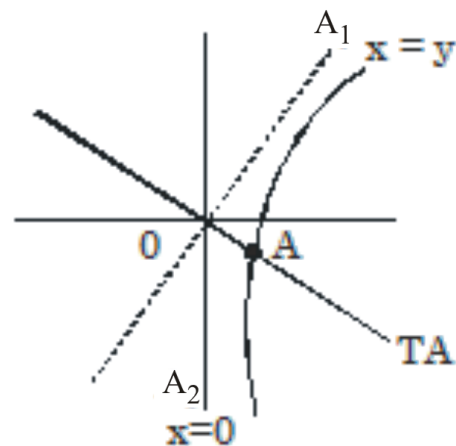
If A'B'C'D' is a square, then tangents  $y = \pm x \pm 5$  for which diagonal length A'B'. Area of circumcircle of  $\Delta A'B'C'$  is  $25\pi$ .

$$\text{Area of circle } C_1 = \frac{25\pi}{2}$$

$$\therefore \text{ Required ratio} = \frac{1}{2}$$

5.

**Ans ( 200.00 )**



$y = x - \frac{1}{x}$  is a hyperbola having asymptotes

$$x^2 - xy = 0 \Rightarrow x = 0 \text{ and } x = y$$

$$\text{equation of TA : } y = \tan\left(-\frac{\pi}{8}\right)x \quad y = (1 - \sqrt{2})x$$

point A is intersection point of TA and H

$$(1 - \sqrt{2})x = x - \frac{1}{x} \Rightarrow x = \frac{1}{2^{1/4}}$$

$$\Rightarrow A\left(\frac{1}{2^{1/4}}, \frac{1 - \sqrt{2}}{2^{1/4}}\right)$$

$$OA = \sqrt{\frac{1}{\sqrt{2}} + \frac{3 - 2\sqrt{2}}{\sqrt{2}}} = \sqrt{2\sqrt{2} - 2}$$

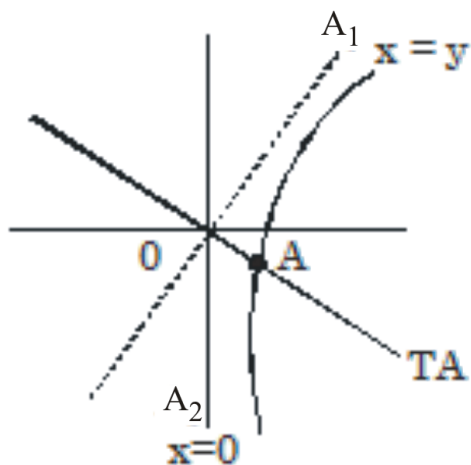
Angle between asymptotes

$$\Rightarrow 2\tan^{-1} \frac{b}{a} = \frac{3\pi}{4}$$

$$\frac{b}{a} = (\sqrt{2} + 1) \Rightarrow \frac{b^2}{a^2} = 3 + 2\sqrt{2} = e^2 - 1$$

$$e = \sqrt{4 + 2\sqrt{2}}$$

6. Ans ( 100.00 )



$y = x - \frac{1}{x}$  is a hyperbola having asymptotes

$$x^2 - xy = 0 \Rightarrow x = 0 \text{ and } x = y$$

$$\text{equation of TA : } y = \tan\left(-\frac{\pi}{8}\right)x$$

$$y = (1 - \sqrt{2})x$$

point A is intersection point of TA and H

$$(1 - \sqrt{2})x = x - \frac{1}{x} \Rightarrow x = \frac{1}{2^{1/4}}$$

$$\Rightarrow A\left(\frac{1}{2^{1/4}}, \frac{1 - \sqrt{2}}{2^{1/4}}\right)$$

$$OA = \sqrt{\frac{1}{\sqrt{2}} + \frac{3 - 2\sqrt{2}}{\sqrt{2}}} = \sqrt{2\sqrt{2} - 2}$$

Angle between asymptotes

$$\Rightarrow 2\tan^{-1} \frac{b}{a} = \frac{3\pi}{4}$$

$$\frac{b}{a} = (\sqrt{2} + 1) \Rightarrow \frac{b^2}{a^2} = 3 + 2\sqrt{2} = e^2 - 1$$

$$e = \sqrt{4 + 2\sqrt{2}}$$

## PART-3 : MATHEMATICS

### SECTION-II (ii)

7. Ans ( 9 )

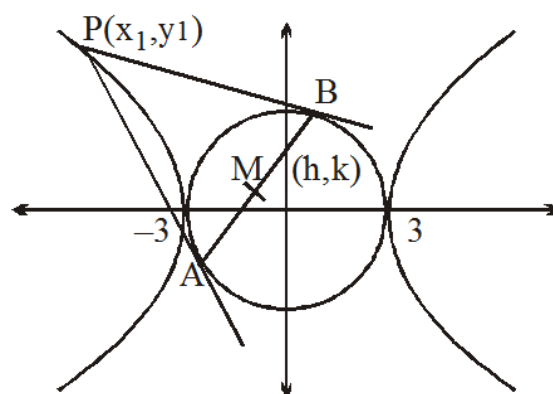
$$H : \frac{x^2}{9} - \frac{y^2}{4} = 1$$

$$\text{Equation of AB: } xx_1 + yy_1 = 9 \quad \dots\dots(1)$$

(Chord of contact)

$$xh + yk = h^2 + k^2 \quad \dots\dots(2)$$

$$(T = S_1)$$



On comparing equation (1) and (2), we get

$$\frac{x_1}{h} = \frac{y_1}{k} = \frac{9}{h^2 + k^2}$$

$$x_1 = \frac{9h}{h^2 + k^2} = 3 \sec \theta$$

$$y_1 = \frac{9k}{h^2 + k^2} = 2 \tan \theta$$

On using  $\sec^2 \theta - \tan^2 \theta = 1$ , we get

$$\left(\frac{3h}{h^2 + k^2}\right)^2 - \left(\frac{9k}{2(h^2 + k^2)}\right)^2 = 1$$

$$\Rightarrow \frac{81x^2}{9} - \frac{81y^2}{4} = (x^2 + y^2)^2$$

$$\Rightarrow \frac{x^2}{9} - \frac{y^2}{4} = \left(\frac{x^2 + y^2}{9}\right)^2$$

$$\therefore k = 9 \text{ Ans.}$$

8. Ans ( 8 )

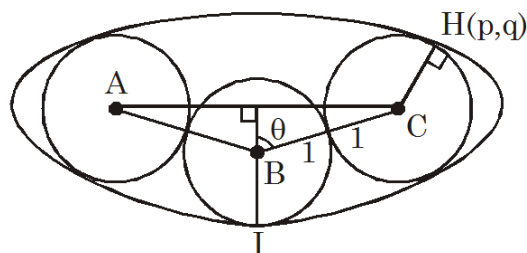
Let  $\angle ABC = 2\theta$ ,  $\angle CBO = \theta$ ,

$OC = 2\sin\theta$ ,  $C \equiv (2\sin\theta, 0)$ .

Let  $H \equiv (p, q)$ ,  $\therefore \frac{p^2}{a^2} + \frac{q^2}{b^2} = 1$  ... (i)

and  $(2\sin\theta - p)^2 + q^2 = 1$  ... (ii)

since slope of CH must be same as slope of normal of the ellipse at H.



$$\Rightarrow \frac{q}{p - 2\sin\theta} = \frac{a^2 q}{b^2 p} \quad \dots (iii)$$

from (i), (ii) and (iii) and using  $\sin^2\theta + \cos^2\theta = 1$

$$\Rightarrow a = b\sqrt{\frac{2}{b-1}}, \text{ area} = \pi ab, A = \pi \cdot b^2 \sqrt{\frac{2}{b-1}}$$

$$\frac{dA}{db} = \sqrt{2}\pi \frac{(\sqrt{b-1})(2b) - b^2 \left(\frac{1}{2\sqrt{b-1}}\right) \cdot 1}{(\sqrt{b-1})^2}$$

$$\Rightarrow \frac{dA}{db} = 0 \Rightarrow (b-1)2b = \frac{b^2}{2}$$

$$\Rightarrow b \neq 0, \Rightarrow 2b - 2 = \frac{b}{2} \quad \& \quad a = \frac{4}{3} \sqrt{\frac{2}{\frac{4}{3} - 1}}$$

$$\Rightarrow 4b - 4 = b$$

$$b = \frac{4}{3} \quad \therefore a = \frac{4}{3} \sqrt{6}$$

Now  $b = 2\cos\theta + 1$

$$\Rightarrow \cos\theta = \frac{1}{6} \Rightarrow \tan\theta = \sqrt{35}$$

$$\Rightarrow \angle AOB = 2\theta = 2\tan^{-1}\sqrt{35}$$

9. Ans ( 329 )

$$(x, y) \text{ --- L --- } \left( \frac{4-3y}{3}, \frac{4-3x}{3} \right)$$

$$\left( \frac{4-3y}{3}, \frac{4-3x}{3} \right)$$

$$\text{--- M --- } \left( \frac{6-3x}{3}, \frac{2-3y}{3} \right)$$

$$\left( 2-x, \frac{2-3y}{3} \right)$$

$$\text{--- N --- } \left( \frac{17-12y}{12}, \frac{5-4x}{4} \right)$$

$$(4 + 4\cos\theta, 3 + 3\sin\theta)$$

$$\text{--- L --- M --- N --- } \left( -\frac{19}{12} - 3\cos\theta, -\frac{11}{4} - 4\sin\theta \right)$$

$\therefore$  equation of ellipse, is

$$\frac{\left(x + \frac{19}{12}\right)^2}{9} + \frac{\left(y + \frac{11}{4}\right)^2}{16} = 1$$

$$\Rightarrow \left(4x + \frac{19}{3}\right)^2 + \left(3y + \frac{33}{4}\right)^2 = 144$$

$$\Rightarrow 16x^2 + 9y^2 + \frac{152}{3}x + \frac{99}{2}y + c = 0$$

$$\therefore \frac{9ag + 152bf}{bg} = \frac{9a}{b} + \frac{152f}{g}$$

$$= \frac{9 \times 16}{9} + \frac{152 \times 99/2}{152/3} = 16 + \frac{297}{2}$$



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(Academic Session : 2022 - 2023)

JEE (Advanced)

PRE - SCORE

14-10-2022

## JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S(TEAS)]

ANSWER KEY

PAPER-2

### PART-1 : PHYSICS

|                 |    |                |       |              |                |       |       |
|-----------------|----|----------------|-------|--------------|----------------|-------|-------|
| SECTION-I (i)   | Q. | 1              | 2     | 3            | 4              | 5     | 6     |
|                 | A. | A,B,C,D        | A,B,D | A,B,D        | A,C            | B,C   | A,C,D |
| SECTION-I (ii)  | Q. | 7              | 8     | 9            | 10             |       |       |
|                 | A. | B              | B     | B            | D              |       |       |
| SECTION-II (i)  | Q. | 1              | 2     | 3            | 4              | 5     | 6     |
|                 | A. | 10.28 to 10.29 | 0.25  | 5.00 to 5.01 | 45.04 to 45.05 | 12.56 | 0.50  |
| SECTION-II (ii) | Q. | 7              | 8     | 9            |                |       |       |
|                 | A. | 5              | 6     | 2            |                |       |       |

### PART-2 : CHEMISTRY

|                 |    |         |         |       |       |       |        |
|-----------------|----|---------|---------|-------|-------|-------|--------|
| SECTION-I (i)   | Q. | 1       | 2       | 3     | 4     | 5     | 6      |
|                 | A. | A,B,C,D | A,B,C,D | A,B,D | B,C,D | A,C,D | A,C    |
| SECTION-I (ii)  | Q. | 7       | 8       | 9     | 10    |       |        |
|                 | A. | D       | B       | B     | B     |       |        |
| SECTION-II (i)  | Q. | 1       | 2       | 3     | 4     | 5     | 6      |
|                 | A. | 11.00   | 4.00    | 3.00  | 53.33 | 19.17 | 178.00 |
| SECTION-II (ii) | Q. | 7       | 8       | 9     |       |       |        |
|                 | A. | 8       | 4       | 3     |       |       |        |

### PART-3 : MATHEMATICS

|                 |    |         |       |      |       |      |      |
|-----------------|----|---------|-------|------|-------|------|------|
| SECTION-I (i)   | Q. | 1       | 2     | 3    | 4     | 5    | 6    |
|                 | A. | A,B,C,D | B,C,D | B,C  | A,B,C | A,D  | B,C  |
| SECTION-I (ii)  | Q. | 7       | 8     | 9    | 10    |      |      |
|                 | A. | B       | A     | C    | D     |      |      |
| SECTION-II (i)  | Q. | 1       | 2     | 3    | 4     | 5    | 6    |
|                 | A. | 6.00    | 9.00  | 8.00 | 16.00 | 2.00 | 1.00 |
| SECTION-II (ii) | Q. | 7       | 8     | 9    |       |      |      |
|                 | A. | 2       | 6     | 11   |       |      |      |

### HINT – SHEET

#### PART-1 : PHYSICS

##### SECTION-I (i)

5. Ans (B,C)

$$\begin{aligned}
 \ell_2 \ell_1 &= \mu_0^2 a_1 a_2 (\ln(8 \times 90) - 2) (\ln \cos 20^\circ - 2) \\
 &= \mu_0^2 - 8 \times 10^{-9} (\ln 320 - 2) \\
 &= 18\pi^2 \times 10^{-19} \times 10^{-9} (0.695 + 3)(3) \\
 M &= \sqrt{128 \times 10^{-17} \times 3 \times 3.7} \\
 &= 10^{-8} \times 8 \times \sqrt{6 \times 3.7} \\
 &= 4 \times 10^{-7} \text{ H}
 \end{aligned}$$

## PART-1 : PHYSICS

### SECTION-I (ii)

10. Ans (D)

(P) If 1<sup>st</sup> case  $E = E_1 = E_0 \sin \omega t$

If 2<sup>nd</sup> case  $E = E_2 = E_0 \sin(\omega t + \phi)$

If 3<sup>rd</sup> case  $E = E_1 + E_2 = E_0' \sin(\omega t + \delta)$

All three cases effective value is same

$$\Rightarrow E_0' = E_0 \Rightarrow \phi = \frac{2\pi}{3}$$

(Q) Case : (2) When  $S_3, S_4$  closed  $z_1 = \sqrt{2}R$

Case : (2) When all opened

$$z_2 = \sqrt{R^2 + (X_C - X_L)^2}$$

$$\Rightarrow X_C - X_L = R \Rightarrow X_C = 2R \Rightarrow \frac{X_C}{R} = 2$$

$$\Rightarrow \frac{4\pi}{5} \frac{X_C}{R} = \frac{8\pi}{5}$$

(R) When  $S_1, S_2$  closed  $Z_1 = \sqrt{5}R$

When  $S_3, S_4$  closed  $Z_2 = \sqrt{2}R$

$$\frac{I_1}{I_2} = \sqrt{\frac{2}{5}} \Rightarrow \pi \left( \frac{I_1}{I_2} \right)^2 = \frac{2\pi}{5}$$

(S) When all opened  $Z_1 = \sqrt{2}R$

When  $S_2, S_3$  closed  $Z_2 = R$

$$\frac{I_3}{I_4} = \frac{1}{\sqrt{2}} \Rightarrow \pi \left( \frac{I_3}{I_4} \right)^2 = \frac{\pi}{2}$$

## PART-1 : PHYSICS

### SECTION-II (i)

6. Ans (0.50)

$$B\ell v + \frac{L di}{dt} - \frac{q}{C} = 0$$

$$B\ell \frac{dv}{dt} = B\ell \frac{dq}{dt}$$

$$v = \frac{B\ell}{m} (Q_0 - q); i = -\frac{dq}{dt}$$

$$\Rightarrow \frac{B^2 \ell^2}{m} (Q_0 - q) \frac{-q}{C} - \frac{L d^2 q}{dt^2} = 0$$

$$\frac{Q_0 - 2q}{C} - \frac{L d^2 q}{dt^2} = 0$$

$$\frac{Q_0 - 2q}{C} = z \Rightarrow \frac{-2}{C} \frac{d^2 q}{dt^2} = \frac{d^2 z}{dt^2}$$

$$\frac{d^2 z}{dt^2} = \frac{-2}{LC} z = -\frac{1}{4} z; \omega = \frac{1}{2} = \frac{2\pi}{T}$$

$$T = 4\pi = 12.56 \text{ Sec}$$

$$\frac{Q_0 - 2q}{C} = z = z_0 \sin(\omega t + \phi)$$

$$\Rightarrow \frac{Q_0}{C} = z_0 \sin \phi,$$

$$-\frac{2}{C} i = z_0 \omega \cos(\omega t + \phi)$$

$$t = 0, i = 0 \Rightarrow \phi = \frac{\pi}{2}; q = \frac{Q_0}{2} (1 - \cos \omega t)$$

$$i = \frac{Q_0 \omega}{2} \sin \omega t; i_{\max} = \frac{10}{2} = 0.5 \text{ mA}$$

## PART-1 : PHYSICS

### SECTION-II (ii)

7. **Ans ( 5 )**

The free-body diagram is drawn for arbitrary positive displacements  $x$  and  $x_B$ . The motion variable  $x$  is measured downward from the position of static equilibrium defined as that which exists when  $x_B = 0$ . The additional stretch in the upper spring, beyond that which exists at static equilibrium, is  $2x - x_B$ .

Therefore, the dynamic spring force in the upper spring, and hence the dynamic tension  $T$  in the cable, is  $k(2x - x_B)$ . Summing forces in the  $x$ -direction gives

$$[\Sigma F_x = m\ddot{x}]$$

$$-2k(2x - x_B) - kx = m\ddot{x}$$

which becomes

$$\ddot{x} + \frac{5k}{m}x = \frac{2kb \cos \omega t}{m}$$

The natural frequency of the system is

$$\omega_n = \sqrt{5k/m}. \text{ Thus,}$$

$$\omega_C = \omega_n = \sqrt{5k/m}$$

8. **Ans ( 6 )**

For the graphs:

$$I = 6A \text{ and } V = t$$

$$\text{Now } \frac{dQ}{dt} = 6$$

$$\int_0^Q dQ = \int_0^t 6dt$$

$$Q = 6t$$

$$\therefore C = \frac{Q}{V} = 6F$$

9. **Ans ( 2 )**

$\phi_{\text{cub}}$  = flux due to single wire from whole cube

$$\phi_{\text{cub}} = \frac{\lambda \ell}{8\epsilon_0} \text{ similarly four wires out of twelve}$$

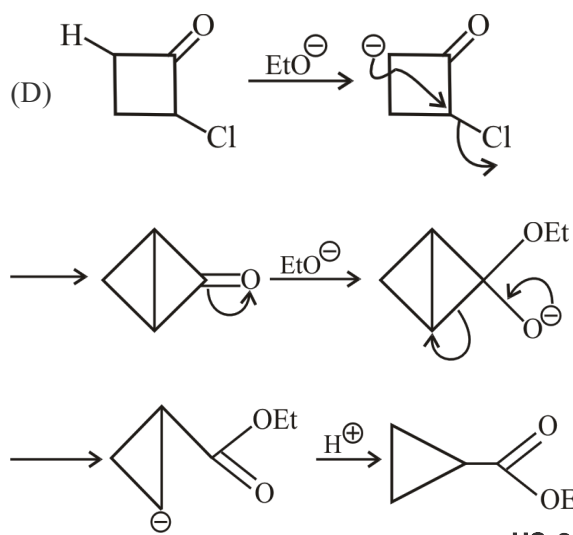
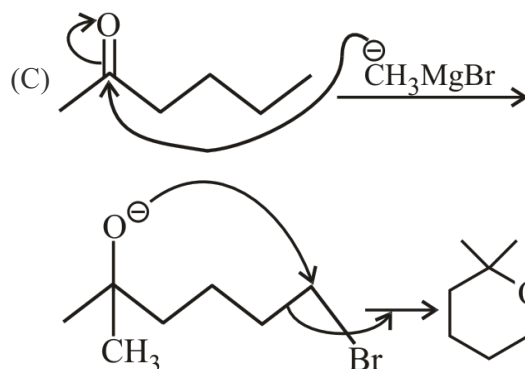
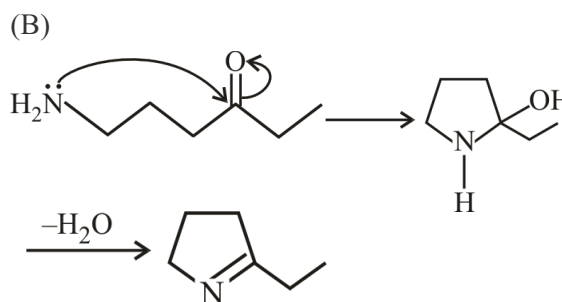
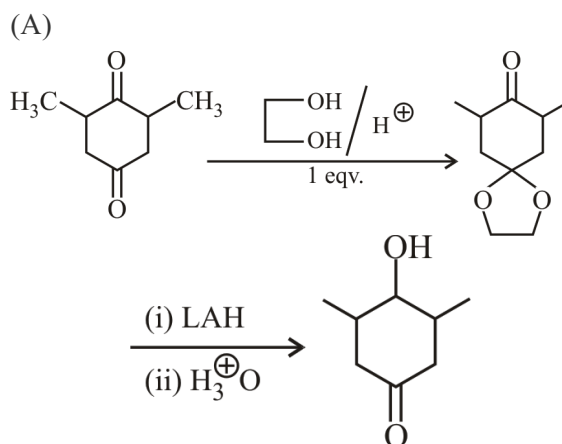
will have same contribution and eight will have zero.

$$\phi_{\text{cub}} = \frac{\lambda \ell}{8\epsilon_0} \times 4 = \frac{\lambda \ell}{2\epsilon_0}$$

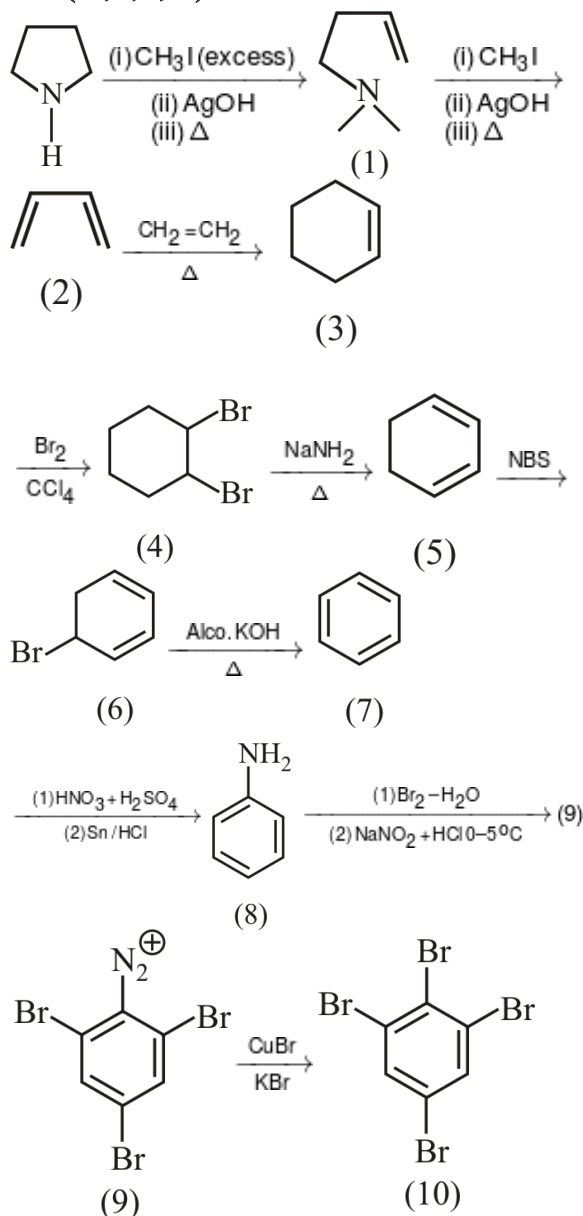
## PART-2 : CHEMISTRY

### SECTION-I (i)

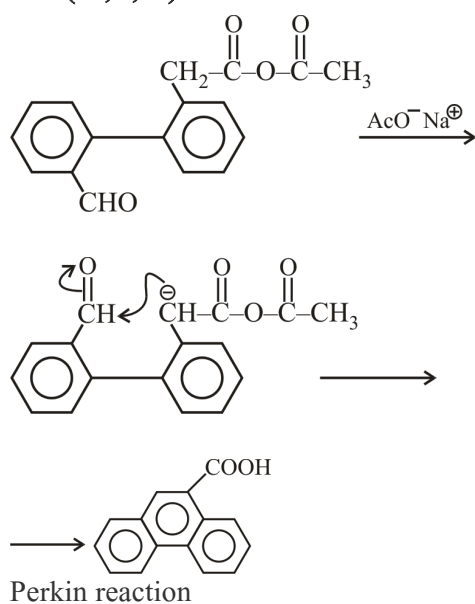
1. **Ans ( A,B,C,D )**



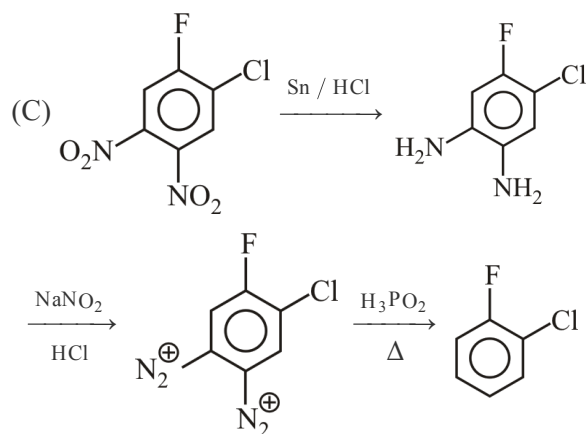
2. Ans (A,B,C,D)



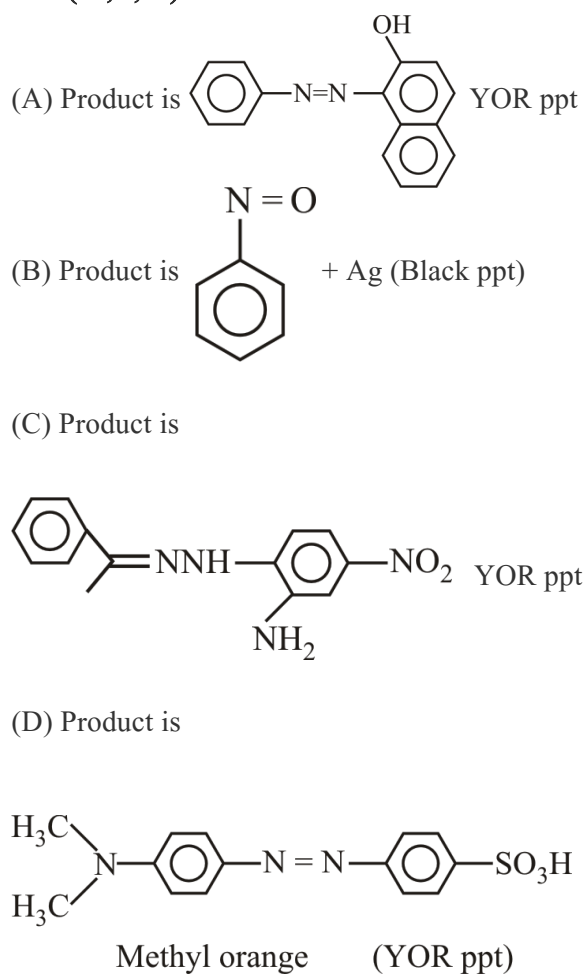
3. Ans (A,B,D)



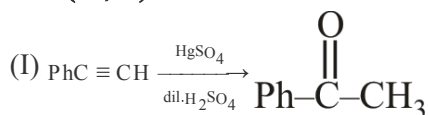
4. Ans (B,C,D)



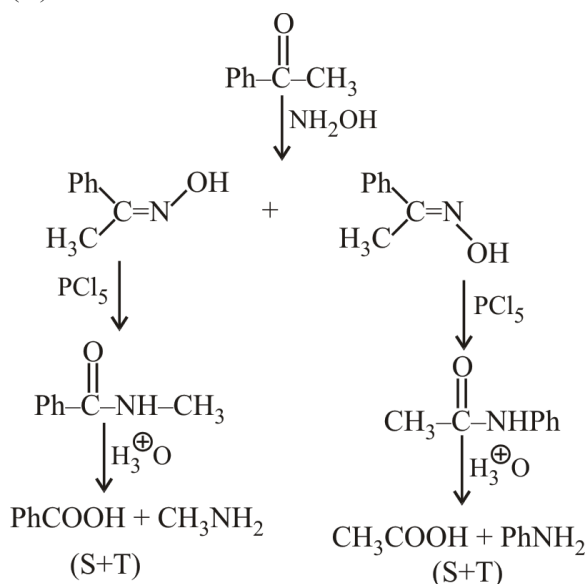
5. Ans (A,C,D)



6. Ans (A,C)



(II)



## PART-2 : CHEMISTRY

### SECTION-I (ii)

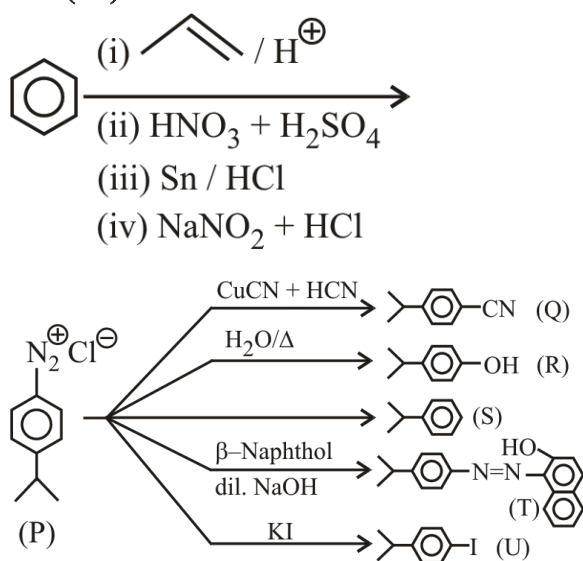
7. Ans (D)

Depends on stability of  $\text{C}^\ominus$

8. Ans (B)

Depends on stability of  $\text{C}^\ominus$

10. Ans (B)

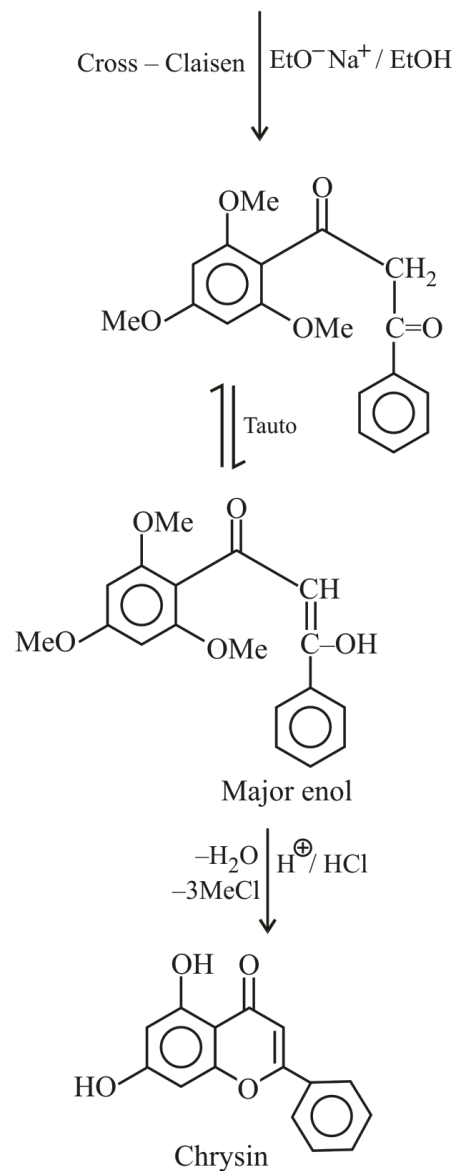
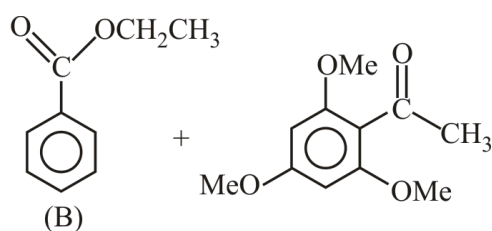
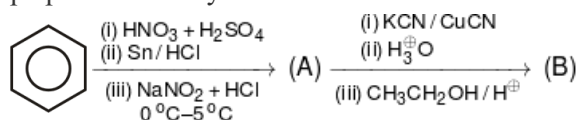


## PART-2 : CHEMISTRY

### SECTION-II (i)

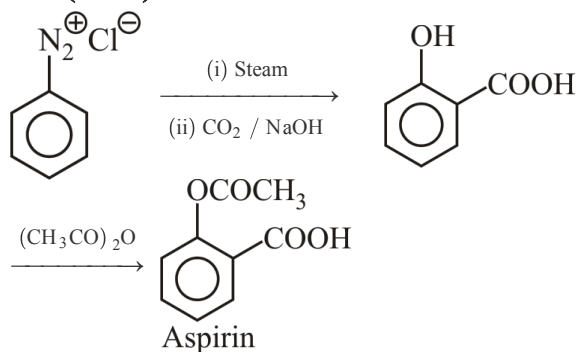
1. Ans (11.00)

Chrysin is an organic compound, found in honey. It is used for bodybuilding for treating anxiety, inflammation, boldness etc. One of the methods of preparation of chrysin follows the reaction scheme :



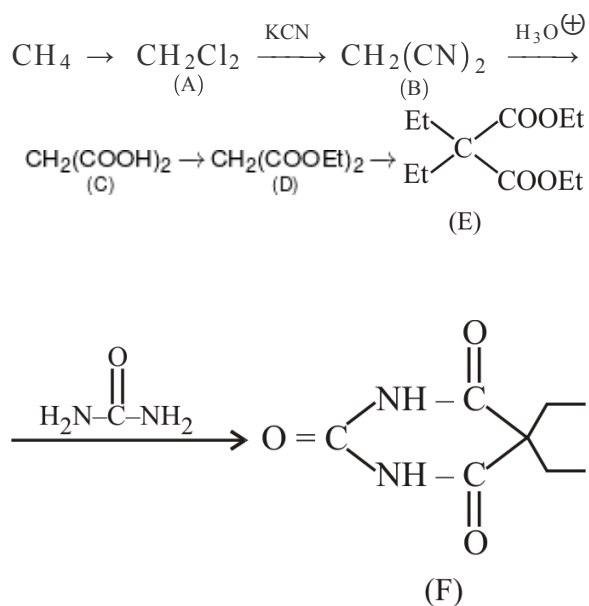
DU = 11

2. Ans ( 4.00 )



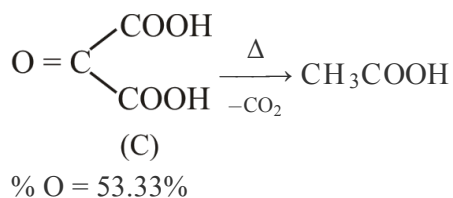
No. of O atoms = 4

3. Ans ( 3.00 )

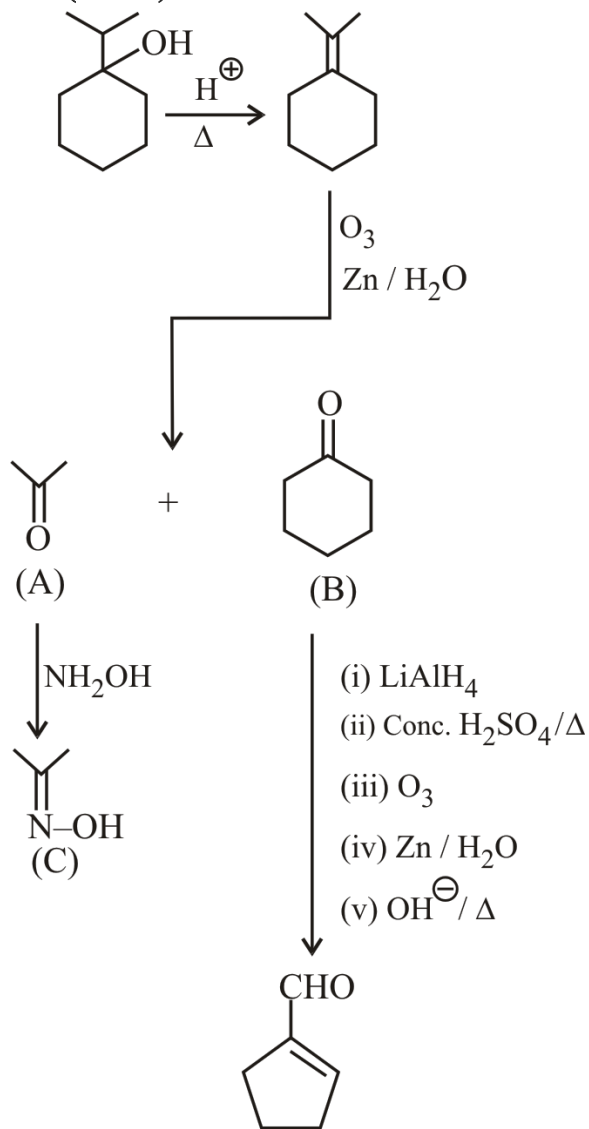


Number of sp<sup>2</sup> c = 3

4. Ans ( 53.33 )

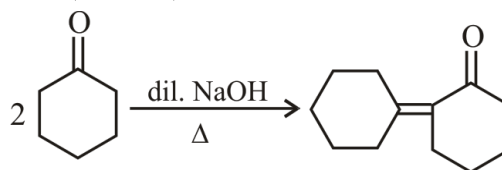


5. Ans ( 19.17 )



% N is (C) = 19.17

6. Ans ( 178.00 )



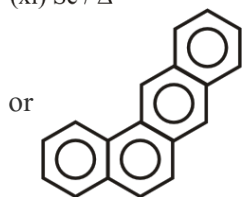
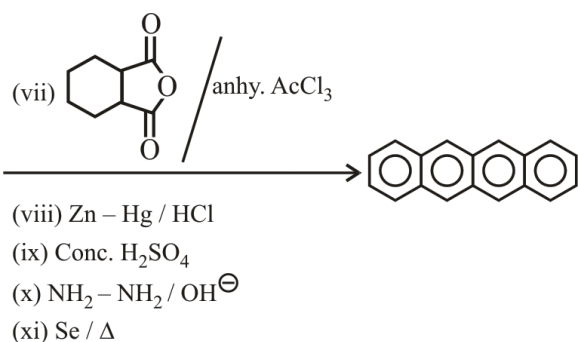
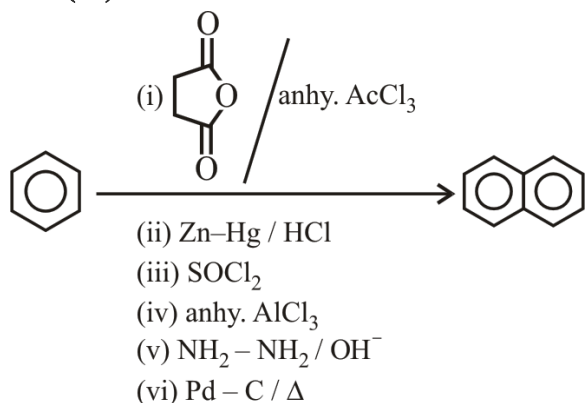
## PART-2 : CHEMISTRY

### SECTION-II (ii)

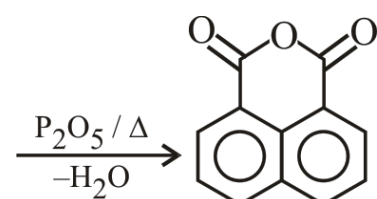
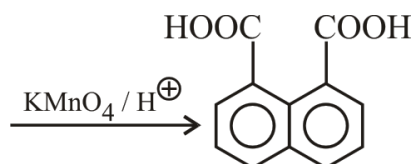
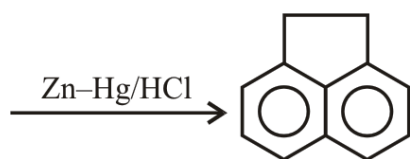
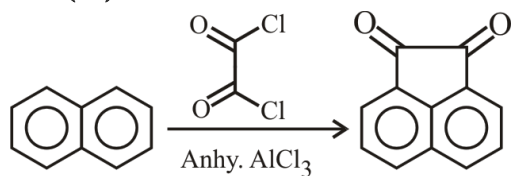
7. Ans ( 8 )

(a), (b), (c), (e), (g), (h), (i), (k) will produce phenol.

8. Ans (4)



9. Ans (3)

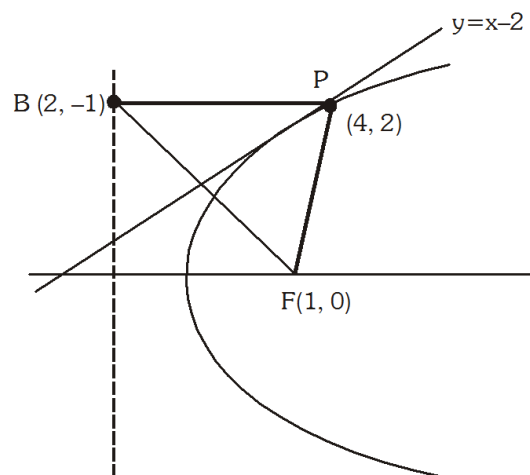


## PART-3 : MATHEMATICS

### SECTION-I (i)

1. Ans (A,B,C,D)

Mirror image B of focus lies on directrix. Also  
 PB || axis of parabola



Point B

$$\frac{x-1}{1} = \frac{y-0}{-1} = \frac{-2(1-0-2)}{(1^2+1^2)}$$

$$\Rightarrow \frac{x-1}{1} = \frac{y}{-1} = 1$$

$$\Rightarrow \begin{cases} x = 2 \\ y = -1 \end{cases}$$

$$\text{Line PB : } (y+1) = \frac{3}{2}(x-2)$$

$$\Rightarrow 3x - 2y = 8$$

Hence equation of axis is  $3x - 2y = \lambda$

satisfying (1, 0) we get  $3x - 2y = 3$

equation of directrix is

$2x + 3y = \mu$  satisfying (2, -1) we get

$$2x + 3y = 1$$

Length of semi-latus rectum

$$= \frac{|2+0-1|}{\sqrt{2^2+3^2}} = \frac{1}{\sqrt{13}}$$

$$\text{Hence length of latus rectum} = \frac{2}{\sqrt{13}}$$

$$SP = PM$$

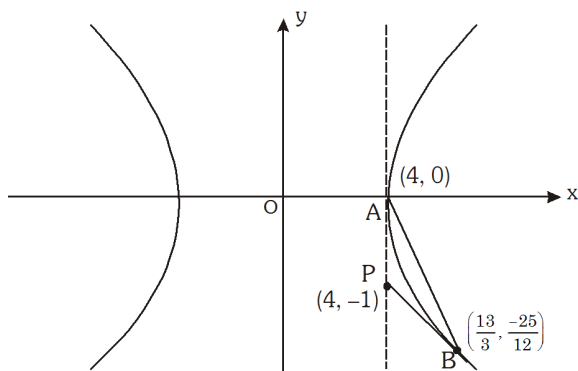
$$\sqrt{(x-1)^2 + y^2} = \left| \frac{2x+3y-1}{\sqrt{2^2+3^2}} \right|$$

$$13(x^2 - 2x + 1 + y^2)$$

$$= 4x^2 + 9y^2 + 1 + 12xy - 4x - 6y$$

$$9x^2 + 4y^2 - 12xy - 22x + 6y + 12 = 0$$

2. Ans (B,C,D)



Hyperbola given is

$$\frac{x^2}{16} - \frac{y^2}{25} = 1$$

Let the slope of tangent = m

equation of tangents PA and PB are

$$y = mx \pm \sqrt{16m^2 - 25}$$

satisfying (4, -1) on the equation of tangent we get

$$-1 = 4m \pm \sqrt{16m^2 - 25}$$

$$\Rightarrow (4m+1)^2 = 16m^2 - 25$$

$$\Rightarrow 16m^2 + 8m + 1 = 16m^2 - 25$$

$$\Rightarrow m = -\frac{13}{4} \text{ and a vertical tangent as } m^2 \text{ is}$$

cancelled

$$\therefore \text{ equation of tangents are } 13x + 4y - 48 = 0$$

and  $x = 4$

points of contacts are (4, 0) and  $\left(\frac{13}{3}, -\frac{25}{12}\right)$

$$\text{by using } \left( \mp \frac{ma^2}{\sqrt{a^2m^2 - b^2}}, \mp \frac{b^2}{\sqrt{a^2m^2 - b^2}} \right)$$

$\therefore$  Area of triangle

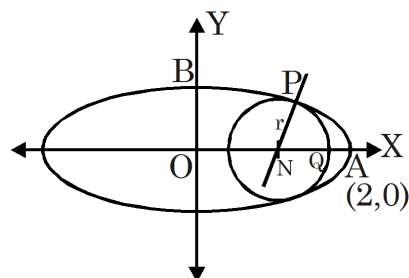
$$\Delta PAB = \frac{1}{2} \times 1 \times \left( \frac{13}{3} - 4 \right) = \frac{1}{6} \text{ sq. unit}$$

$\therefore$  options B, C, D are correct

3. Ans (B,C)

$$\frac{x^2}{4} + \frac{y^2}{3} = 1$$

$$N \equiv (k, 0) \text{ \& } Q(r+k, 0)$$



Normal PN at P  $(2 \cos \theta, \sqrt{3} \sin \theta)$ ;  $\theta \neq \frac{n\pi}{2}$  is

$$2x \sec \theta - \sqrt{3}y \operatorname{cosec} \theta = 1$$

$$\Rightarrow N \equiv \left( \frac{\cos \theta}{2}, 0 \right) \Rightarrow k = \frac{\cos \theta}{2} \in \left( 0, \frac{1}{2} \right)$$

$$[\text{as } k > 0 \text{ \& } \theta \neq \frac{n\pi}{2}]$$

So, all such circles touches the arc of the ellipse in the 1<sup>st</sup> (as well as 4<sup>th</sup>) quadrant between (strictly) A & B.

$$\text{Thus, } 0 < k < \frac{1}{2} \text{ \& } r+k < 2 \quad (\text{as } Q_x < A_x)$$

Now, these is another possibility when point of normalcy is A (note it cannot be B otherwise  $k=0$ )

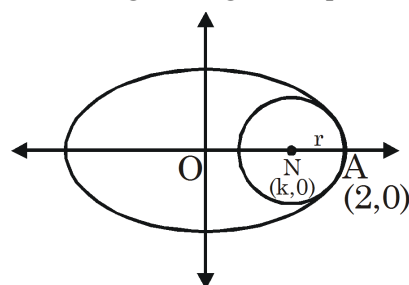
In this case,

$$r+k=2$$

So, circle is

$$(x-k)^2 + y^2 = (2-k)^2$$

Intersecting it with given ellipse :



$$3x^2 + 4[(2-k)^2 - (x-k)^2] = 12$$

$$\Rightarrow x^2 - 8kx + 16k - 4 = 0$$

$$\Rightarrow x = 2, 8k - 2$$

Since, the circle lies completely inside the ellipse

So, other point of intersection

(i.e.  $x = 8k - 2$ ) should be imaginary.

Hence,  $8k - 2 > 2$  (Note  $8k - 2 > -2$  as  $k > 0$ )

$$\Rightarrow k > \frac{1}{2} \text{ \& } r+k=2$$

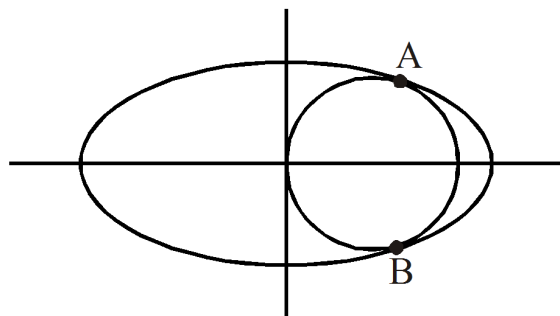
4. Ans (A,B,C)

E and C touch each other at A and B

$$\text{solve E and C} \equiv b^2x^2 + a^2[c^2 - (x-c)^2] = a^2b^2$$

$$\Rightarrow b^2x^2 + a^2[2cx - x^2] = a^2b^2$$

$$\Rightarrow (b^2 - a^2)x^2 + 2ca^2x - a^2b^2 = 0$$



$$D = 0 \Rightarrow 4c^2a^4 = 4a^2b^2(a^2 - b^2)$$

$$\Rightarrow a^2 = \frac{b^4}{b^2 - c^2}$$

$$\Delta = \pi ab = \frac{\pi b^3}{\sqrt{b^2 - c^2}} = \frac{\pi}{\sqrt{b^{-4} - c^2b^{-6}}}$$

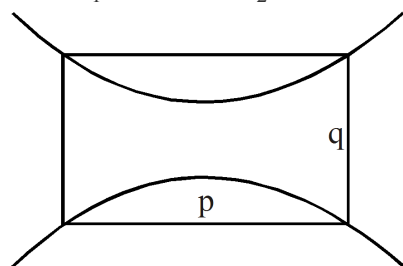
$$\frac{d\Delta}{db} = 0 \Rightarrow 6b^{-7}c^2 = 4b^{-5} \Rightarrow b^2 = \frac{3c^2}{2}$$

$$b^2 = \frac{3c^2}{2} \text{ and } a^2 = \frac{9c^2}{2}$$

$$\therefore e = \frac{\sqrt{2}}{\sqrt{3}}, \Delta = \frac{3\sqrt{3}\pi}{2}c^2$$

5. Ans (A,D)

$$p = \frac{2b_1^2}{a_1} \text{ and } q = \frac{2b_2^2}{a_2}$$



$$\Rightarrow p = 2a_1(e_1^2 - 1) \text{ and } q = 2a_2(e_2^2 - 1)$$

$$\text{Also } 2a_1e_1 = q \text{ and } 2a_2e_2 = p$$

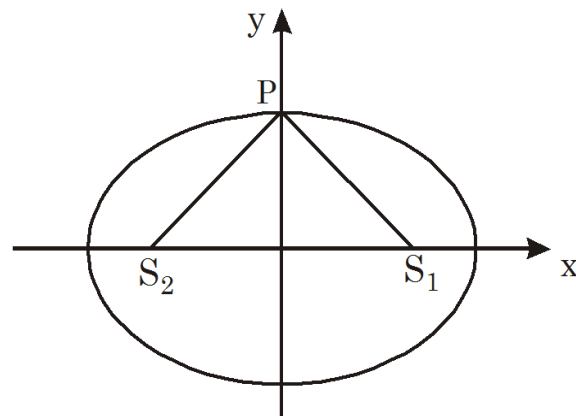
$$\Rightarrow \frac{p}{q} = \frac{e_1^2 - 1}{e_1} \text{ and } \frac{q}{p} = \frac{e_2^2 - 1}{e_2}$$

$$\Rightarrow \left(e_1 - \frac{1}{e_1}\right) \left(e_2 - \frac{1}{e_2}\right) = 1$$

$$\Rightarrow e_1e_2 + \frac{1}{e_1e_2} = 1 + \frac{e_1}{e_2} + \frac{e_2}{e_1}$$

6. Ans (B,C)

$$\frac{x^2}{16} + \frac{y^2}{7} = 1$$



$$S_1S_2 = 2ae = 2\sqrt{a^2 - b^2}$$

$$= 2\sqrt{16 - 7} = 6$$

$$\text{Given } \Delta = 3\sqrt{7}$$

$$\therefore \frac{1}{2} \times h \times 6 = 3\sqrt{7}$$

$$h = \sqrt{7}$$

$\therefore$  P must lie on minor-axis

$$\text{Now } S_1P = S_2P = \sqrt{9 + 7} = 4$$

$$\text{Semiperimeter} = S_1P + S_2P + S_1S_2$$

$$= \frac{2 \times 4 + 6}{2} = \frac{14}{2} = 7$$

**Ex-radii of Circles**

(i) Opposite to P

$$r_1 = \frac{\Delta}{\text{semiperimeter} - S_1S_2}$$

$$r_1 = \frac{3\sqrt{7}}{7 - 6} = 3\sqrt{7}$$

(ii) Opposite to  $S_1$ (or  $S_2$ )

$$r_2 \text{ (or } r_3) = \frac{\Delta}{\text{semiperimeter} - S_1P}$$

$$= \frac{3\sqrt{7}}{7 - 4} = \sqrt{7}$$

(Since  $S_1PS_2$  is an isosceles triangle hence ex-radii opposite to  $S_1$  and  $S_2$  are equal)

## PART-3 : MATHEMATICS

### SECTION-I (ii)

#### 7. Ans (B)

Equation of line AB :  $y = 2x + 1$

Any point on AB is  $(\lambda, 2\lambda + 1)$

equation of chord of contact of  $(\lambda, 2\lambda + 1)$  with respect to parabola  $y^2 = 4x$ .

$$y \cdot (2\lambda + 1) = 2 \cdot (x + \lambda)$$

$$(2x - y) + \lambda(2 - 2y) = 0.$$

$$y = 1, x = \frac{1}{2}$$

$$\therefore (x, y) = \left(\frac{1}{2}, 1\right).$$

#### 8. Ans (A)

Solving  $y = 2x + 1$

with  $y = x^2$

$$A = (1 + \sqrt{2}, (1 + \sqrt{2})^2), B = ((1 - \sqrt{2}), (1 - \sqrt{2})^2)$$

$$\therefore AB = \sqrt{(2\sqrt{2})^2 + (4\sqrt{2})^2} = 2\sqrt{10}$$

length of perpendicular from  $(0, 0)$  to AB =  $\frac{1}{\sqrt{5}}$ .

$$\therefore \text{area}(\Delta AOB) = \frac{1}{2} \cdot 2\sqrt{10} \cdot \frac{1}{\sqrt{5}} = \sqrt{2}$$

#### 9. Ans (C)

$\therefore$  The portion of tangent intercepted between asymptotes is bisected at the point of contact.

$$\Rightarrow AM = BM = 3\sqrt{3}.$$

$\therefore$  co-ordinates of A and B are

$$= (5 \pm 3\sqrt{3} \cos \theta, \sqrt{3} \pm 3\sqrt{3} \sin \theta)$$

$$\text{where } \tan \theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$$

Let

$$A = \left(5 + 3\sqrt{3} \cdot \frac{\sqrt{3}}{2}, \sqrt{3} + 3\sqrt{3} \cdot \frac{1}{2}\right) = \left(\frac{19}{2}, \frac{5\sqrt{3}}{2}\right)$$

$$B = \left(5 - 3\sqrt{3} \cdot \frac{\sqrt{3}}{2}, \sqrt{3} - 3\sqrt{3} \cdot \frac{1}{2}\right) = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$$

$\therefore$  Equation of asymptotes (OA and OB)

$$y = \frac{5\sqrt{3}}{19}x \text{ and } y = -\sqrt{3}x.$$

combined equation

$$(19y - 5\sqrt{3}x) \cdot (y + \sqrt{3}x) = 0$$

$$19y^2 + 14\sqrt{3}xy - 15x^2 = 0$$

#### 10. Ans (D)

$$m_{OA} = \frac{5\sqrt{3}}{19}, m_{OB} = -\sqrt{3}$$

$$\therefore \tan \theta = \left| \frac{\frac{5\sqrt{3}}{19} + \sqrt{3}}{1 - \frac{5\sqrt{3}}{19} \cdot \sqrt{3}} \right| = \left| \frac{24\sqrt{3}}{4} \right| = 6\sqrt{3}$$

$$\Rightarrow \theta = \tan^{-1} 6\sqrt{3} = \cos^{-1} \frac{1}{\sqrt{109}}$$

## PART-3 : MATHEMATICS

### SECTION-II (i)

#### 1. Ans (6.00)

If  $\alpha = 8$ , then tangent to parabola is

$$y = mx + \frac{2}{m} \text{ \& this to be tangent to ellipse}$$

$$\frac{4}{m^2} = 32m^2 + 8$$

$$\Rightarrow 8m^4 + 2m^2 - 1 = 0$$

$$m^2 = \frac{-2 \pm \sqrt{4 + 32}}{16} = \frac{-2 \pm 6}{16} = \frac{1}{4}, -\frac{1}{2}$$

(Rejected)

$$\Rightarrow m = \pm \frac{1}{2} \Rightarrow x - 2y + 8 = 0$$

#### 2. Ans (9.00)

Equation of normal to parabola  $y^2 = 8x$ .

$$y = mx - 4m - 2m^3$$

$\therefore$  It passes through  $(a, 0)$

$$\therefore 2m^3 + 4m - am = 0$$

$$m - 0 \text{ \& } 2m^2 + 4 - a = 0$$

$$\Rightarrow m^2 = \frac{a - 4}{2} \Rightarrow m = \pm \sqrt{\frac{a - 4}{2}}$$

Now, two of them are perpendicular

$$\therefore \left(\frac{a - 4}{2}\right) = 1 \Rightarrow a = 6$$

$\therefore (a, 0)$  is on ellipse

$$\therefore \frac{a^2}{4a} = 1 \Rightarrow a = \frac{36}{4} = 9$$

3. Ans ( 8.00 )

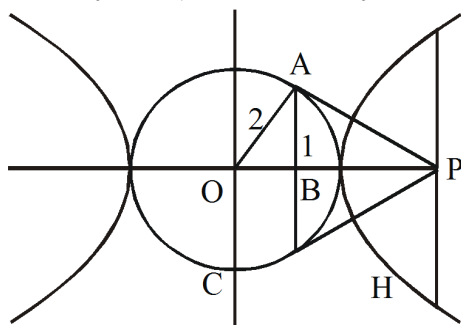
$$\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1 \quad \therefore e_1 = e_2$$

$$\Rightarrow e_1 = \sqrt{2} \Rightarrow a = b$$

foot of perpendicular drawn from focus upon any tangent always lies on auxiliary circle.

$$x^2 + y^2 = a^2 \Rightarrow a = 2 \Rightarrow b = 2$$

$$C : x^2 + y^2 = 4 \text{ (which is auxiliary circle itself)}$$



If ellipse & hyperbola cuts orthogonally  $\Rightarrow$  confocal.

$$\text{foci of H} \Rightarrow (\pm 2\sqrt{2}, 0) \Rightarrow (4e_e, 0) = (2\sqrt{2}, 0)$$

$$\Rightarrow e_e = \frac{1}{\sqrt{2}}$$

$$\beta^2 = 16 \left( 1 - \frac{1}{2} \right) = 8 \Rightarrow \beta = 2\sqrt{2}$$

4. Ans ( 16.00 )

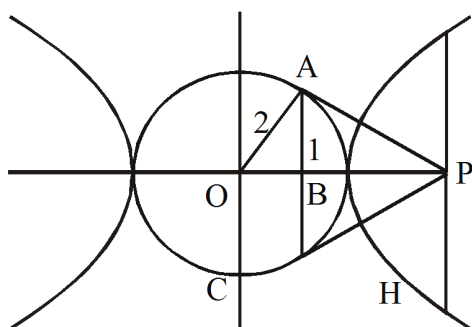
$$\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1 \quad \therefore e_1 = e_2$$

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foot of perpendicular drawn from focus upon any tangent always lies on auxiliary circle.

$$x^2 + y^2 = a^2 \Rightarrow a = 2 \Rightarrow b = 2$$

$$C : x^2 + y^2 = 4 \text{ (which is auxiliary circle itself)}$$



$\angle BOA = 30^\circ$  which is eccentric angle of P

$\Rightarrow$  Length of double ordinate is

$$2b \tan 30^\circ = 2 \cdot 2 \cdot \frac{1}{\sqrt{3}} = \frac{4}{\sqrt{3}}$$

5. Ans ( 2.00 )

$$\text{Given } \frac{dy}{dx} = \frac{k}{y} \Rightarrow y dy = k \cdot dx$$

$$\frac{y^2}{2} = kx + c \text{ passes through } (0,0)$$

$$\Rightarrow c = 0$$

$$\therefore y^2 = 2kx \text{ passes through } (1,2)$$

$$\text{curve is } y^2 = 4x.$$

Equation of tangent to  $y^2 = 4x$  at  $P(1,2)$  is

$$y = x + 1.$$

$\therefore$  equation of required circle

$$(x-1)^2 + (y-2)^2 + \lambda(y-x-1) = 0$$

$$\text{passes through } (1,0) \Rightarrow \lambda = 2$$

$$\therefore \text{circle is } (x-1)^2 + (y-2)^2 + 2(y-x-1) = 0$$

$$\Rightarrow x^2 + y^2 - 4x + 2y + 3 = 0$$

$$\therefore \text{radius} = \sqrt{4+1-3} = \sqrt{2}$$

6. Ans ( 1.00 )

Normal to parabola  $y^2 = 4x$  is

$$y = mx - 2m - m^3 \text{ passes through } (3,-1)$$

$$\Rightarrow m^3 - m - 1 = 0 ; \text{ roots are } m_1, m_2, m_3$$

$$\text{Let } \frac{1-m}{1+m} = x \Rightarrow m = \frac{1-x}{1+x}$$

Put value of  $m$  in (1)

$$\therefore (1) \Rightarrow \left( \frac{1-x}{1+x} \right)^3 - \left( \frac{1-x}{1+x} \right) - 1 = 0$$

$$\Rightarrow x^3 - x^2 + 7x + 1 = 0$$

$$\Rightarrow \text{sum of roots} = 1$$

## PART-3 : MATHEMATICS

### SECTION-II (ii)

7. Ans (2)

Any point on  $(t+2)(x+y)=1$  is  $\left(\alpha, \frac{1}{t+2} - \alpha\right)$

Equation of chord of contact is

$$4x\alpha + 16y\left(\frac{1}{t+2} - \alpha\right) = 1$$

$$\Rightarrow \alpha(4x - 16y) + \left(\frac{16}{t+2}y - 1\right) = 0$$

the chord of contact passes through the intersection

$$\text{of } x - 4y = 0 \text{ and } \left(\frac{16}{t+2}y - 1\right) = 0$$

$$\therefore \text{the point of intersection is } \left(\frac{t+2}{4}, \frac{t+2}{16}\right)$$

It lies inside the ellipse if

$$4\left(\frac{t+2}{4}\right)^2 + 16\left(\frac{t+2}{16}\right)^2 - 1 < 0$$

$$= \left(-\frac{4}{\sqrt{5}} - 2\right) < t < \left(\frac{4}{\sqrt{5}} - 2\right) \Rightarrow t = -3, -1.$$

8. Ans (6)

Given equation,  $ax + y\sqrt{\alpha^2 - 1} = 1$

$$\text{or } y = -\frac{\alpha}{\sqrt{\alpha^2 - 1}}x + \frac{1}{\sqrt{\alpha^2 - 1}}.$$

$$\therefore c^2 = a^2m^2 - b^2$$

$$\frac{1}{\alpha^2 - 1} = \frac{a^2 \cdot \alpha^2}{\alpha^2 - 1} - b^2$$

$$1 = a^2\alpha^2 - b^2(\alpha^2 - 1)$$

$$\therefore \alpha^2(a^2 - b^2) + (b^2 - 1) = 0. \text{ holds for all}$$

permissible values of  $\alpha$ .

$\Rightarrow$  equation is an identity

$$\Rightarrow a^2 - b^2 = 0 \text{ and } b^2 = 1$$

$$\therefore a = b = 1.$$

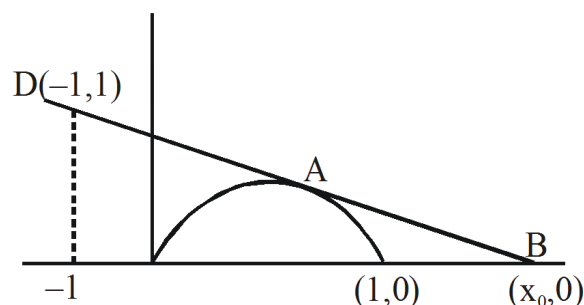
$$\text{Now } e^2 = \frac{1 + b^2}{a^2} = 1 + 1 \Rightarrow e = \sqrt{2}$$

$$\therefore [e^5] = [4\sqrt{2}] = 5.$$

9.

Ans (11)

$$\left(\frac{2\sqrt{3}-3}{3} \Rightarrow 11\right)$$



We need to find  $x_0$  such that line determinant

by

$(-1, 1)$  &  $(x_0, r)$  is tangent to parabola

$$\text{equation of PA } \Rightarrow (y - 1) = m(x + 1)$$

$$x - x^2 - 1 = mx + m$$

$$x^2 + (m - 1)x + m + 1 = 0$$

$$D = 0 \quad m = \frac{6 \pm \sqrt{36 + 12}}{2}$$

$$(m - 1)^2 - 4(m + 1) = 0 \quad m = 3 - \sqrt{12}$$

$$m^2 - 6m - 3 = 0$$

$$\text{eq. of PA } \rightarrow y - 1 = (3 - \sqrt{12})(x + 1)$$

$$y = 0 \Rightarrow x + 1 = \frac{1}{\sqrt{12} - 3} = \frac{\sqrt{12} + 3}{3}$$

$$x_0 = \frac{\sqrt{12}}{3} = \frac{2\sqrt{3}}{3}$$

$$\text{Closest the receiver can be } \frac{2\sqrt{3}}{3} - 1 = \frac{2\sqrt{3} - 3}{3}$$